

Overfitting Intuition

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Choose Informative Questions

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Types of Classification Error

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Human Classifier Simulation

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Interpret a gap for Overfitting and model capacity. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Overfitting and model capacity. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply independently

Models A training 4 validation 5, B training 1 validation 2, C training 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Rule A memorises eight training IDs and gets 8/8 training, 2/8 validation. Rule B uses genre and gets 6/8 training, 6/8 validation. Which is preferable on this evidence, and what test remains?

6. Solve the given case

Rule A memorises six training book names. It scores 6/6 familiar and 2/6 new. Rule B uses relevant traits but is untested.

What does the evidence support?

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Rule A memorises six training book IDs. Rule B uses a relevant topic feature. Why might A fail on a new ID? What test is needed?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 16 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Avoid title-by-title patches; improve features or thresholds that transfer.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Rule A memorises eight training IDs and gets 8/8 training, 2/8 validation. Rule B uses genre and gets 6/8 training, 6/8 validation. Which is preferable on this evidence, and what test remains?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Select from evidence

Choose among the three worked candidates and explain the training trap.

My reasoning and evidence

13. W2 Interpret a gap

Does a training-validation gap alone prove overfitting?

My reasoning and evidence

14. W3 Propose a controlled change

Describe one fair next development comparison.

My reasoning and evidence

15. W4 Apply independently

Models A training 4 validation 5, B training 1 validation 2, C training 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

My reasoning and evidence

16. W5 Limit a diagnosis

What extra evidence would you inspect before concluding a model memorized noise?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Overfitting Intuition

A fictional project compares three book-length prediction models on the same development sets.

Mean absolute errors in pages: Simple training 5, validation 6; Medium training 2, validation 3; Complex training 0, validation 7.

Select lowest validation error; untouched final tests are not used for selection.

Training and validation errors

	Training MAE pages	Validation MAE pages
Simple	5	6
Medium	2	3
Complex	0	7

Read a cell using both its row and column labels.

Overfitting Intuition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training comparison

Complex fits training perfectly with zero error.



2. Validation comparison

Medium has error 3, versus Simple 6 and Complex 7.



3. Choose for development

Select Medium under the declared lowest-validation-error rule.



4. Final evaluation

Freeze the selection, then evaluate new reserved cases.

CHANGE ONE THING

Complex's validation error becomes 2 on a newly documented, fair development comparison.

Explain the changed result using the matching lesson.

Does zero training error justify choosing Complex in the original record?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Compare recommender revisions on fixed development cases, reserving new cases for the final presentation.